

# Early Predictors of Medical Intractability in Childhood Epilepsy

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The goal of this study is identify early predictors of intractability in childhood epilepsy. A cohort of epileptic children living in the northwest sector of Hong Kong was prospectively identified and monitored. Treatment effect was analyzed at the last follow-up before July 1, 2000. Cases were patients who had an average of at least one unprovoked seizure per month during an observational period of at least 2 years. Controls were children having achieved at least 2 seizure-free years. Strong univariate association was observed between intractability and the following factors: high initial seizure frequency, remote symptomatic etiology, infantile spasms and mixed seizure types, abnormal neurologic status, history of status epilepticus, neonatal seizures, and early breakthrough attacks after treatment initiation. Independent predictors of intractability with multiple regression were abnormal neurodevelopmental status, symptomatic etiology, and more than three seizures in the second 6 months after treatment. Our study suggested that risk of developing intractable epilepsy might be predicted, to some extent, at the early course of illness in children with abnormal neurologic status and lack of early response to treatment. © 2003 by Elsevier Inc. All rights reserved.

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## Introduction

In spite of advances in the treatment of epilepsy, a significant number of children with epilepsy do not respond well to antiepileptic drugs (AED). Traditionally, studies of seizure outcome have focused on remission of seizures. This wealth of information comes in part because total seizure control is relatively easy to define. Intracta-

bility of epilepsy is not, however, the converse of seizure freedom. In fact, the definition of drug resistance varies. Studies of remission do not directly address development of intractability. Relatively few studies have specifically addressed risk factors associated with medical refractoriness and its predictors [1-4]. Besides, there are no published data of intractable childhood epilepsy in a Chinese population. Early identification of patients who are at high risk of developing intractable epilepsy would be essential in parental counseling and selecting patients for more intensive investigations and treatment, for example, early consideration of epilepsy surgery.

This study formed part of an epidemiologic survey of childhood epilepsy in the northwest sector of Hong Kong. In this area, the prevalence rate of active epilepsy for children under 15 years of age was 1.52 in 1,000 on July 1, 1997 [5]. From the cohort of patients identified, a follow-up study was carried out to identify early predictors of intractability in childhood epilepsy, by comparing children suffering from intractable epilepsy and those with well-controlled epilepsy.

## Methods

We recruited children with active epilepsy who were under 15 years of age and who had attended the Neurodevelopmental and Pediatric Clinics in Tuen Mun Hospital before July 1, 1997. Electroencephalogram (EEG) referrals and hospital admissions for seizures were also reviewed for patient selection. Tuen Mun Hospital is the only hospital that serves the northwest sector of Hong Kong. Patients were referred from various sources including Maternal and Child Health Clinics, Student Health Care, general practitioners and private pediatricians, Accident and Emergency Departments, and special schools. In Hong Kong, epileptic children are usually under the care of hospital-based pediatric neurologists. General practitioners, private pediatricians, and adult neurologists do not treat epileptic children. Therefore it would be safe to assume our study population represents an unbiased pediatric epileptic population. EEG (both awake and asleep tracings were obtained whenever possible), computed tomography (CT), magnetic resonance imaging (MRI), chromosomal analysis, urine test screening, and neuropsychologic assessment were performed when needed for a precise diagnosis. Patients and their

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families were interviewed (K.L.K) when they were initially recruited into the survey. The following data were obtained: details of seizures, age of seizure manifestation, family history of seizure disorders, history of febrile seizure, status epilepticus (SE), neonatal seizures (NS), etiologic factors, additional neuroimpairments, EEG at first visit, and neuroimaging findings. We analyzed treatment effect at the last follow-up before July 1, 2000. Epilepsy diaries were reviewed and patients were interviewed for clarification of seizure status and drug compliance.

### Inclusion and Exclusion Criteria

Only children who were residents of the study area and who had two or more unprovoked seizures were included. Children with neonatal seizures were included only if the seizures had ceased by the time of the neonatal discharge from the hospital and later recurred without provocation or continued after the late neonatal period. Children with only febrile seizures, acute symptomatic seizures, isolated seizure, and neonatal seizures were excluded.

### Definitions

Definitions and principles of the International League Against Epilepsy (ILAE) Commission on Classification and Terminology and the Commission on Epidemiology and Prognosis were used [6-8]. Classification of the seizure types was made according to the revised International Classification of Epileptic Seizures (ICES) [6], with additional categories for infantile spasms and mixed seizure types (to allow separate classification of children with multiple seizure types, with none being dominant). The classification was based on seizure descriptions and EEG. Etiology was defined according to the epidemiologic standards set forth by the ILAE [8]. "Remote symptomatic" meant either that the child had preexisting neurologic abnormalities, a history of brain insult, or a disorder associated with an increased risk of epilepsy that was presumed to be etiologically related to the child's epilepsy.

Patients with intractable epilepsy were included in the study as cases. To meet criteria of intractable epilepsy, a child had to have uncontrolled seizures that occurred with an average frequency of at least one seizure per month over an observational period of 2 years or more, despite treatment with at least three different AEDs, administered singly or in combination [1,9]. Drugs had to be pushed to maximum tolerated doses and non-compliance ruled out by medical review and interview with parents. Children with poor compliance or low serum drug levels were excluded. An AED was considered to have failed if it did not control seizures, or if the medication was discontinued because of adverse effects. An inappropriate drug use, for example, carbamazepine for absence epilepsy, was not considered as drug failure. Controls were patients with well-controlled epilepsy, which was defined as having achieved at least 2 seizure-free years before last follow-up.

### Data Analysis

Data were analyzed with chi-square and the chi-square test for trend for univariate comparison. Odds ratios (OR) and 95% confidence interval (CI) were calculated. Forward logistic regression was used for multivariate analysis.

### Results

A total of 309 children were prospectively identified, out of which 44 (14.2%) met the study definition of intractable epilepsy and 211 (68.3%) were in the well-controlled comparison group. Table 1 depicts characteristics of the two groups of children. Mean age at seizure onset was 2.65 years for cases and 5.953 years for controls

**Table 1. Characteristics of case and control groups**

| Parameters                     | Cases<br>(n = 44) | Controls<br>(n = 211) |
|--------------------------------|-------------------|-----------------------|
| Duration of follow-up (years)  |                   |                       |
| Mean                           | 5.88              | 4.67                  |
| Median                         | 4.5               | 4                     |
| Male (%)                       | 61.4              | 83.9                  |
| Median number of different AED | 2                 | 1                     |
| Polytherapy (%)                | 61.3              | 1.9                   |
| Current seizure frequency (%)  |                   |                       |
| 1-2/month                      | 20.5              | 0                     |
| 3-6/month                      | 22.7              | 0                     |
| 3-4/week                       | 22.7              | 0                     |
| Daily                          | 34.1              | 0                     |
| Deaths                         | 0                 | 0                     |

( $P < 0.001$ ). None of our patients died during the follow-up period. Seizure frequency in cases was one to two seizures per month in 20.5%, three to six per month in 22.7%, three to four per week in 22.7%, and daily in 34.1%.

### Profile of Intractable Epilepsy

Among the 44 patients with intractable epilepsy, the most common seizure types were partial seizures (54.5%), infantile spasms (22.7%), and mixed seizure types (11.4%). Forty (91%) children had remote symptomatic etiology. Etiology could be established in 72.5% patients with remote symptomatic etiology: prenatal/perinatal in 55% and postnatal in 41%. The prenatal/perinatal causes included the following: central nervous system (CNS) malformations (3), perinatal hypoxic-ischemic encephalopathy (8), prematurity (2), small for gestation (2), and congenital infection (1). Postnatal causes were as follows: meningitis/encephalitis (6), encephalopathy (1), trauma (2), mesial temporal sclerosis (3), and others (1). First EEG was abnormal in 85.4% of cases with focal abnormality in 51%. Neuroimaging was performed in 84% of cases with abnormalities detected in 64.9%: focal infarct (7), generalized encephalomalacia (4), neuronal migration defect (3), mesial temporal sclerosis (3), and others (8).

### Analysis of Factors Related to Seizure Outcome—Full Group

Results of univariate analysis on the relationship between predictive risk factors and seizure intractability are demonstrated in Table 2. Age at onset  $<1$  year was a significant predictor of intractability (OR = 4.3, 95% CI 1.85, 10.03). The association appeared throughout the whole age range. Daily seizures before therapy were statistically associated with the development of intractable epilepsy. More than 20 seizures before medication and intractability were strongly associated (OR = 4.57, 95% CI 2.16, 9.7). Infantile spasms and mixed seizures were more common in the cases than the control group. Syn-

**Table 2. Univariate comparison of case and control groups**

| Factors                                                             | Cases<br>(n = 44) | Controls<br>(n = 211) | Odds<br>Ratio | 95% Confidence<br>Interval | P Value            |
|---------------------------------------------------------------------|-------------------|-----------------------|---------------|----------------------------|--------------------|
| Age at seizure onset (years) (continuous variable)                  |                   |                       | 0.79          | 0.71, 0.88                 | <0.001             |
| Daily seizures at onset (%)                                         | 59                | 22.7                  | 7.21          | 3.36, 15.46                | <0.001             |
| More than 20 seizures before therapy (%)                            | 36.4              | 10.9                  | 4.57          | 2.16, 9.7                  | <0.001             |
| Initial seizure types (%)                                           |                   |                       |               |                            | <0.001<br>(df = 3) |
| Partial                                                             | 54.5              | 45.4                  |               |                            |                    |
| Generalized                                                         | 11.4              | 50.2                  |               |                            |                    |
| Infantile spasms                                                    | 22.7              | 2.8                   |               |                            |                    |
| Mixed                                                               | 11.4              | 1.4                   |               |                            |                    |
| Epilepsy syndrome (%)                                               |                   |                       |               |                            | <0.001<br>(df = 5) |
| Localization-related                                                |                   |                       |               |                            |                    |
| Idiopathic                                                          | 2.3               | 11.3                  |               |                            |                    |
| Cryptogenic                                                         | 6.8               | 20.9                  |               |                            |                    |
| Symptomatic                                                         | 45.5              | 13.3                  |               |                            |                    |
| Generalized                                                         |                   |                       |               |                            |                    |
| Idiopathic                                                          | 2.3               | 41.2                  |               |                            |                    |
| Cryptogenic/symptomatic                                             | 36.4              | 13.3                  |               |                            |                    |
| Undetermined                                                        | 6.8               | 0                     |               |                            |                    |
| Etiology (%)                                                        |                   |                       |               |                            |                    |
| Remote symptomatic                                                  | 91                | 27.3                  | 26.38         | 9.03, 76.92                | <0.001             |
| Neurodevelopmental status (%)                                       |                   |                       |               |                            |                    |
| Abnormal                                                            | 88.6              | 5.4                   | 22.67         | 8.5, 60.60                 | <0.001             |
| Status epilepticus (%)                                              | 22.7              | 7.6                   | 3.58          | 1.5, 8.54                  | <0.001             |
| Neonatal seizures (%)                                               | 13.6              | 1.9                   | 8.17          | 2.2, 30.30                 | <0.001             |
| Febrile seizures (%)                                                | 18.2              | 16.1                  | 1.16          | 0.5, 1.48                  | 0.74               |
| First electroencephalogram (%)                                      |                   |                       |               |                            |                    |
| Abnormal                                                            | 85.4              | 64.1                  | 2.31          | 1, 5.27                    | 0.047              |
| Neuroimaging (%) (cases 37, controls 133)                           |                   |                       |               |                            |                    |
| Abnormal                                                            | 64.9              | 15.8                  | 3.8           | 3.03, 44.44                | 0.009              |
| ≥3 breakthrough seizures in the first 6 months after treatment (%)  | 81.8              | 33.8                  | 14.83         | 6.63, 33.22                | <0.001             |
| ≥3 breakthrough seizures in the second 6 months after treatment (%) | 81.8              | 31.3                  | 34.48         | 15.15, 81.3                | <0.001             |

dromic grouping and etiology were strongly associated with intractability. The risk was highest in the symptomatic localization-related group, 20 of 48 (41.7%), intermediate in the symptomatic/cryptogenic generalized group, 16 of 44 (36.4%), and lowest in the idiopathic generalized group, 2 of 113 (1.8%). Abnormal neurologic status (mental retardation and/or cerebral palsy) was detected in 88.6% of cases and 5.4% of controls. The occurrence of SE was significantly more common in cases than controls. Status epilepticus occurred in 22.1% of remote symptomatic epilepsy, 8.5% of cryptogenic epilepsy, and 1.8% of idiopathic epilepsy ( $P < 0.05$ ). Neonatal seizures, abnormal first EEG, and abnormal neuroimaging findings were observed more often in cases than control groups. Poor response to therapy was associated with intractable epilepsy; more than three breakthrough seizures either in the first or second 6 months after initiation of treatment were more common in the intractable than well-controlled group.

Multivariate analyses were performed to assess the association between intractable epilepsy and potential independent predictive factors (Table 3). In the overall group, abnormal neurodevelopmental status (OR 18.16, 95% CI 5.19, 63.61), daily seizures before therapy (OR

1.58, 95% CI 1, 2.51), ≥3 breakthrough seizures in the second 6 months after treatment (OR 21.86, 95% CI 7.35, 65.01), and febrile seizure (OR 4.83, 95% CI 1.31, 17.83) were associated with intractable epilepsy.

**Table 3. Multivariate analysis for predictors of intractable epilepsy**

| Factors Retained in Model                                       | Odds<br>Ratio | 95% Confidence<br>Interval | P Value |
|-----------------------------------------------------------------|---------------|----------------------------|---------|
| Abnormal neurodevelopmental status                              | 18.16         | 5.19, 63.61                | <0.001  |
| Daily seizure frequency                                         | 1.58          | 1, 2.51                    | 0.05    |
| ≥3 breakthrough seizures in the second 6 months after treatment | 21.86         | 7.35, 65.01                | <0.001  |
| Febrile seizure                                                 | 4.83          | 1.31, 17.83                | 0.03    |
| Predictability 93%                                              |               |                            |         |

**Table 4. Univariate comparison of case and control groups—without infantile spasms**

| Factors                                                             | Cases<br>(n = 34) | Controls<br>(n = 205) | Odds<br>Ratio | 95% Confidence<br>Interval | P Value            |
|---------------------------------------------------------------------|-------------------|-----------------------|---------------|----------------------------|--------------------|
| Age at seizure onset (years) (continuous variable)                  |                   |                       | 0.96          | 0.88, 1.04                 | 0.325              |
| Daily seizures at onset (%)                                         | 47.1              | 21                    | 3.45          | 1.62, 7.33                 | 0.001              |
| More than 20 seizures before therapy (%)                            | 20.6              | 9                     | 2.64          | 1, 01, 6.9                 | 0.048              |
| Initial seizure types (%)                                           |                   |                       |               |                            | <0.001<br>(df = 3) |
| Partial                                                             | 70.6              | 46.8                  |               |                            |                    |
| Generalized                                                         | 14.7              | 52.7                  |               |                            |                    |
| Mixed                                                               | 14.7              | 0.5                   |               |                            |                    |
| Epilepsy syndrome (%)                                               |                   |                       |               |                            | 0.01<br>(df = 5)   |
| Localization-related                                                |                   |                       |               |                            |                    |
| Idiopathic                                                          | 2.9               | 11.7                  |               |                            |                    |
| Cryptogenic                                                         | 8.8               | 21.5                  |               |                            |                    |
| Symptomatic                                                         | 58.8              | 13.7                  |               |                            |                    |
| Generalized                                                         |                   |                       |               |                            |                    |
| Idiopathic                                                          | 2.9               | 42.4                  |               |                            |                    |
| Cryptogenic/symptomatic                                             | 17.6              | 10.7                  |               |                            |                    |
| Undetermined                                                        | 8.8               | 0                     |               |                            |                    |
| Etiology (%)                                                        |                   |                       |               |                            |                    |
| Remote symptomatic                                                  | 88.2              | 26.3                  | 21            | 7.06, 62.26                | <0.001             |
| Neurodevelopmental status (%)                                       |                   |                       |               |                            |                    |
| Abnormal                                                            | 85.3              | 24.4                  | 17.98         | 6.61, 48.93                | <0.001             |
| Status epilepticus (%)                                              | 20.6              | 7.8                   | 3.06          | 1.15, 8.12                 | 0.025              |
| Neonatal seizures (%)                                               | 11.8              | 2                     | 6.7           | 1.59, 20.22                | 0.01               |
| Febrile seizures (%)                                                | 16.6              | 23.3                  | 1.55          | 0.65, 3.71                 | 0.327              |
| First electroencephalogram (%)                                      |                   |                       |               |                            |                    |
| Abnormal                                                            | 74.2              | 63                    | 1.69          | 0.72, 3.98                 | 0.22               |
| Neuroimaging (%) (cases 29, controls 123)                           |                   |                       |               |                            |                    |
| Abnormal                                                            | 44.9              | 10.6                  | 6.88          | 2.71, 17.44                | 0.001              |
| ≥3 breakthrough seizures in the first 6 months after treatment (%)  | 79.4              | 19.4                  | 16.02         | 6.5, 39.48                 | <0.001             |
| ≥3 breakthrough seizures in the second 6 months after treatment (%) | 73.3              | 10.4                  | 23.81         | 9.82, 57.74                | <0.001             |

1.58, 95% CI 1, 2.51), more than 3 seizures in the second 6 months after treatment (OR 21.86, 95% CI 7.35, 65.01), and febrile seizures (OR 4.83, 95% CI 1.31, 17.83). After adjustment for these factors, factors that were significant on univariate analysis but no longer significant on multivariate analysis included: seizure types, etiology, epilepsy syndromes, neonatal seizures, status epilepticus, abnormal first EEG, and abnormal neuroimaging study.

#### **Analysis of Factors Related to Seizure Outcome—Without Infantile Spasms**

Separate models were constructed for children who did not have infantile spasms. Table 4 demonstrates the results of univariate analysis on the relationship between predictive risk factors and intractability. Daily seizures, more than 20 seizures before medication, mixed seizure types, syndromic grouping, etiology, abnormal neurodevelopmental status, status epilepticus, neonatal seizure, abnormal neuroimaging, and more than three breakthrough seizures either in the first or second 6 months after treatment were all associated with bad seizure outcome. Age at seizure onset and abnormal first EEG were no longer significant when children with infantile spasms were excluded from the analysis.

Independent predictors of intractability with multiple regression were: symptomatic etiology (OR 16.58, CI 4.2, 65.43,  $P = 0.001$ ) and more than 3 seizures in the second 6 months after treatment (OR 11.34, CI 3.53, 36.4,  $P < 0.001$ ).

#### **Discussion**

Early identification of children who are at high risk of developing intractable epilepsy is important. Parents can be properly counseled. Surgically treatable cases could be selected early in their clinical course for intensive investigations. Toxicity from an overdose of antiepileptic drugs and useless polytherapy in medically intractable patients could be minimized. Few studies have examined early predictors of intractability in childhood epilepsy. To the best of our knowledge, study on seizure outcome in a Chinese population is largely unknown. Although the present study is hospital-based, we believe the study population was representative of the population considered. There is no private pediatric neurologist in the area we studied. General practitioners and pediatricians usually do not manage children with epilepsy. Electroencephalogram records are considered a comprehensive source of patient identification [10,11]. We focused on information

prospectively collected before any knowledge of seizure outcome was available, thus bias in ascertaining history of various risk factors in cases and controls could be minimized.

Approximately 14% of our children with active epilepsy experienced medically intractable seizures. According to a multi-institutional study on intractable childhood epilepsy in Japan, the frequency of intractable epilepsy is 13% [12]. Sillanpaa, in an analysis of risk factors in a cohort of 178 children, reported drug-refractory epilepsy in 22% of their patients [13]. Ten percent of the cohort reported by Berg et al. met the criteria of intractability early in the course of their disorders [4]. The discrepancy in frequency can be explained by the true differences in various socioeconomic and geographical situations. The variability also reflects the lack of a standard and well-accepted definition of intractable epilepsy. Our definition addresses three fundamental factors: medication failure, seizure frequency, and time of observation [1-4,9,14]. Not all children who failed to enter remission would be considered intractable according to our definition. The three most salient early predictors of seizure refractoriness identified in the present study were high initial seizure frequency, abnormal neurodevelopmental status, and breakthrough seizures early in the course of treatment. Febrile seizures were significant on multivariate analysis but not on univariate analysis.

In accordance with other authors, we observed that age at onset was an influential predictor on univariate analysis [1-3,15]. This finding appeared throughout whole age range. The prognosis was progressively better with increasing age at onset. However, the association was not retained on multivariate analysis and when we excluded those children with infantile spasms. Early age of onset was observed frequently in the remote symptomatic group and those who were neurologically abnormal. Berg et al. also reported that the predictive value of age appeared throughout the whole age range even after the authors controlled for infantile spasms in the analysis [1]. In Casetta's study, the association between age at onset and intractable epilepsy was not completely confirmed throughout the age range [3]. Another study that investigated intractable epilepsy in a population of mentally retarded children also failed to demonstrate persistence of seizures and age at onset [16]. The controversy could be partly explained by the differences between community-based and hospital-based studies. The present study is hospital-based with age at onset of both case and controls comparable to that of Berg's series [1].

Initial seizure frequency was identified to be a significant prognostic factor in our study. Daily frequency before therapy was more commonly observed in cases than in controls. The statistical significance was retained in the multivariate analysis for the full group. Few studies addressed the prognostic significance of seizure frequency. Berg et al. found an effect over a continuum of seizure frequency [4]. Sillanpaa identified children with epilepsy in the catchment area of Turku University Hos-

pital in Finland and followed survivors for 30 years; he noted high frequency of initial seizures as one of the predictors of persistent epilepsy on multivariate analysis [17]. Seizure frequency (<1 versus >1 seizure/month) has been examined in studies of remission and relapse after stopping medication [18,19]. Shinnar et al. reported that in children with two seizures, an interval of less than 6 months between the seizures was predictive of having more than 10 seizures [20].

Patients with more than 20 seizures before treatment were associated with intractable epilepsy in univariate analysis. Infantile spasms and partial seizures accounted for 56% and 19% of our cases, respectively. The association was retained even after children with infantile spasms were eliminated from data analysis. All except one patient had remote symptomatic etiology. Camfield et al. reported that <20 seizures before initiation of treatment was a significant predictor of seizure remission [21]. Others have noted that having many seizures before treatment predicts relapse when medication is withdrawn after 2 to 5 seizure-free years [22-24].

There is limited published data that directly addressed the importance of seizure control and intractable epilepsy. We observed an association between three or more breakthrough seizures in the second 6 months after initiation of treatment and seizure refractoriness. The association was retained also in multivariate analysis. In remission studies, the duration of active epilepsy before achieving control has been thought to be the most important clinical predictor of remission [25,26]. If seizures are not controlled in the first year after diagnosis, only about 60% of patients can be expected to achieve remission [25,27,28]. Occurrence of many generalized seizures before control was one of the predictors of relapse after medication withdrawal [29,30]. Arts et al. found that not achieving a 3-month remission at 6 months had a significant impact on the prognosis [27]. These findings raise the question whether it is the epileptic disposition of the patient making him refractory to treatment or the persistence of seizures making seizures worse.

There was an association between status epilepticus and seizure intractability in univariate analysis but it was no longer retained in the multivariate analysis. Seventy-eight percent of SE in our patients belonged to the remote symptomatic group. In more recent studies of outcome after SE, mortality, morbidity, and seizure recurrence were almost entirely related to the underlying etiology [31,32]. SE should be considered as a marker rather than a cause of poor control.

Abnormal neurologic status was identified to be a significant predictor of seizure intractability in the present study, both in the univariate and multivariate analysis for the full group. Remote symptomatic etiology was a significant independent predictor only when children with infantile spasms were excluded. Regarding underlying causes, CNS infection and perinatal hypoxic-ischemic encephalopathy were the most important underlying dis-

orders in our patients with intractable epilepsy. In Casetta's series of intractable epilepsy, perinatal brain injuries predominated [3], while the most characteristic finding of underlying causes in intractable localization-related epilepsy was CNS infection in Ohtsuka's study [15]. In the acute stage of CNS infection and perinatal hypoxic-ischemic encephalopathy, patients often experienced repeated SE or frequent seizures requiring intensive care. Seizures in the acute stage often originate from several different regions in the individual patient, and multifocal seizures continue after the acute stage. These patients are often resistant to AED.

We observed that febrile convulsion was an independent factor of seizure intractability in full group analysis. The association between febrile seizures and seizure intractability is controversial. Berg et al. observed a non-significant, slightly protective association between febrile seizures and intractability [4]. There are studies that did not find febrile seizures to be associated with seizure outcome [1,27], while Camfield et al. noted that prolonged febrile seizures were associated with intractable epilepsy [33]. The association between febrile seizures and intractable epilepsy seen in surgical series is not a simple cause-effect relationship. A good explanation of our finding is lacking.

Since most pediatric neurologists consider children with infantile spasms relatively intractable, separate models were constructed for those children who did not have infantile spasms. The findings were similar to the full group analysis on univariate comparison. Factors that were no longer significant on univariate comparison were age at seizure onset and abnormal first EEG. Independent risk factors for intractability were symptomatic etiology and more than three seizures in the second 6 months of treatment. The significance of daily seizures was not retained on multiple regression when infantile spasms were excluded. The association between lack of early response to treatment and intractability remained throughout the analysis.

The concept of medical intractability is ill defined. Not being in remission is not necessarily the same as being refractory to treatment. Failure to enter into remission may be attributable to factors including not being treated, poor drug compliance, and relapse after drug discontinuation. Neurologic abnormalities, complex partial or secondarily generalized epilepsy, infantile spasms, remote symptomatic etiology, a large number of seizures, high seizure frequency, and a history of SE seem to be consistent predictors of reduced probability of remission [21,25,34-36]. However, these factors lack specificity for identification of intractability at the time of diagnosis. Our study suggests that risk of developing intractable epilepsy may be predicted, to some extent, at the early course of illness in children with abnormal neurologic status, symptomatic etiology, and especially in those who showed a lack of early response to treatment.

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